ERWIN POUSSI

Robotics & Autonomy | Robot Learning ▪ Scene Understanding ▪ Loco-Manipulation
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EDUCATION

Stanford University | Stanford, CA
PhD in Robotics GPA: 4.1/4.0
Coursework: *Robot Autonomy; Decision Making under Uncertainty; Optimal & Learning-Based Control; Deep RL.*

École Polytechnique | Palaiseau, France
M.Eng. in Computer Science & Mechanical Engineering; Autonomy concentration; Entrepreneurship minor. GPA: 3.93/4.0

WORK & RESEARCH EXPERIENCE

Robotics Researcher — Robot Learning & Autonomy 2026–Present
Stanford University | Multi-Robot Systems Lab & Autonomous Systems Lab

- Generated synthetic data using RRT-planned trajectories in simulated 3D Gaussian Splatting environments and extended SINGER with goal-conditioned imitation learning (DAgger), improving sample efficiency and raising navigation success **from 52% to 90%**. view work
- Developed GAIT for navigation toward language-specified targets, combining persistent scene memory with vision-language reasoning and motion planning; improved success while reducing reasoning calls and latency in controlled GOAT-Bench comparisons; working on sim-to-real transfer. view work
- Developing agentic robotics systems linking scene memory, grasping and whole-body motion planning for **G1 and Spot** robots through DARPA TIAMAT, working on sim-to-real transfer for locomotion and manipulation. view work

AI Researcher — Reinforcement Learning & Scientific Agents Oct. 2025–Present
Stanford School of Medicine | Qiu Lab

- Formulated gene-panel selection as a sequential decision problem and developed **actor–critic RL** to refine gene combinations; on renal carcinoma data, a **500-gene panel** outperformed conventional selectors in clustering agreement while preserving cell-type structure comparable to using 20,000+ genes. view work
- Led research in 4D embryo alignment for PantheonOS, coupling optimal transport with spatial registration to map cells across developmental stages; improved alignment consistency by 21% on embryonic heart data against a reference set of allowed cell-fate transitions. view work
- Developed a generative virtual embryo model that reconstructs tissue anatomy from a single computational seed cell using neural cellular automata and a neural decoder; contributing to the **Virtual Embryo Challenge**. Virtual Embryo Challenge

Machine Learning & Simulation Intern Apr.–Aug. 2025
Stevens Institute of Technology | Computational Fluid Dynamics

- Ran 80,000+ flow simulations and combined physics-informed surrogate modeling with optimization across simulation fidelities for Mars-landing parachute fabric; optimized pore geometries reproduced experimental permeability with **<1% mean error**. view work

Multiphysics Simulation Intern Jun.–Sep. 2024
Daher Aerospace

- Developed and calibrated electromagnetic–thermal simulations of inductive composite welding in ANSYS, matching experimental temperatures to support control of heating and final part geometry.

Autonomous Docking — Research Project Sep. 2023–May 2024
École Polytechnique / The Exploration Company

- Developed vision-based relative navigation and guidance for **Nyx's autonomous docking**; tested closed-loop maneuvers using a quadcopter as the hardware platform. view work

Officer Cadet — AI Tool Development Nov. 2022–Apr. 2023
French Police Cybercrime Unit

- Led trainee software engineers developing an AI tool to analyze leaked communications with police investigators; documented methods for new unit members.

PUBLICATIONS

Under review: Ghasimi, S. D., **Poussi, E.**, Rabinovitch, J. "Surrogate-Based Optimization of Broadcloth Fabric Permeability for Mars Landing Parachute Simulations." Revised manuscript submitted for final review, 2026.
Preprint: Xu, W., **Poussi, E.**, et al. "PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery." bioRxiv, 2026. doi:10.64898/2026.02.26.707870

TECHNICAL SKILLS & LANGUAGES

Python, C/C++, PyTorch, ROS2, MuJoCo, ANSYS; reinforcement learning, imitation learning, perception, motion planning and control.
Languages: French (native), English (TOEFL 109), Mooré (fluent), German (basic).

LEADERSHIP & DISTINCTIONS

Stanford BGSA Fundraising Chair: raised $3,000+ toward a $10,000+ team effort. **Inspirit AI** Instructor: taught hands-on machine learning and neural networks (2026).
Cheer Up: supported young cancer patients. **Mentoring:** mathematics, physics and university-application guidance for students in Burkina Faso, France and the U.S.
French Government Excellence Fellowship (2022); **1st nationwide** in Burkina Faso's mathematics & physical sciences baccalauréat (2020).